



Islamic University of Technology

Department of Computer Science and Engineering (CSE)

CSE 4206: Digital Logic Design Lab
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Lab Final
Instructions

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1 Hardware Tasks

- You will be given a Boolean expression on a piece of paper. You have to simplify the expression and implement the simplified expression using **basic gates on a breadboard**.
- You will be given approximately **10 minutes** to implement the circuit. After that, you will need to fill in the **truth table** based on the output of the implemented circuit.
- You have to write your ID in the given piece of paper and submit it when done.
- You have to perform the tasks individually.
- **If you fill in the truth table without implementing the circuit, you will not receive any marks for it.**
- Clear your breadboard once you are done with the task. If you don't there will be a penalty of **5 marks**.

2 Software Tasks

- You will be given a piece of paper with a task to be performed in Proteus. Once you receive it, check that your name and ID are correctly mentioned. You must perform the task assigned to you.
- You will be given approximately **45 minutes** to implement the combinational circuit in Proteus. After your evaluation, you need to submit the file to Google Classroom.
- You are allowed to connect to the internet to check Google Classroom posts and instructions. However, you are **not allowed to search for the solution** to the task assigned to you. You must implement the circuit on your own.
- If you have any issues regarding the task, you may only ask the course teacher for assistance.
- You **must not take help from anyone**. If you are found taking help from someone, you will receive a warning. If you are found doing so again, you will be **EXPULSED FROM THE LAB FINAL**.
- You **must not reuse any previous files or anyone else's files**. If you are found doing so, it will result in **EXPULSION FROM THE LAB FINAL**.
- If you have successfully implemented the circuit but fail to explain its operation, it will result in a **severe penalty**.